BIO-RAD

Gene Pulser® Electroprotocol

Cell Type Fungal / Yeast

Species Used *Dictyostelium discoideum*

Molecules Electroporated DNA: supercoiled and linear

Before the Pulse

Cell Growth Medium HL5 (peptone, yeast extract, glucose)

(ATCC Media #671)

Growth Phase at Harvest 1 x 10 (6) to 1 x 10 (7) cells / ml

Pre-pulse Incubation 5 min.

Wash Solution Hepes Buffered Saline

The Pulse

Instruments Used Gene Pulser® apparatus & Pulse Controller

Electroporation Temperature 4 °C

Electroporation Medium Hepes Buffered Saline

Cell Density 5 x 10 (6) / ml

Volume of Cells 1 ml

DNA Concentration 20 ng to 20 µg

DNA Resuspension Buffer TE (10 mM Tris, 1 mM EDTA, pH 8.0)

Volume of DNA Any volume

Cuvette Gap 0.4 cm

Voltage 1.25 kV

Field Strength 3.125 kV/cm

Capacitor 25 µF

Resistor (Pulse Controller) Ω none. NOT

Time Constant 0 .5 to 0.7 msec

After the Pulse

Outgrowth Medium HL5

Outgrowth Temperature 22 °C

Length of Incubation overnight

Selection Method or Assay Used G418

Electroporation Efficiency 10 (-3) transformants/ cell; 6 x 10 (3) transformants / µg DNA.

Per Cent Survival >90%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

**It is NOT RECOMMENDED to use high voltage with out the Pulse Controller.

HBS: 10mM HEPES, pH 7.2,150 mM NaCl, 5 mM CaCl2

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